Li Ju

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Education

Ph.D., Scientific Computing — Sep. 2021 – Jun. 2026
Uppsala University, Uppsala, Sweden

M.Sc., Computational Science — Sep. 2019 – Aug. 2021
Uppsala University, Uppsala, Sweden

M.Sc., Chemometrics, *with Honors* — Sep. 2016 – Jun. 2019
University of Science and Technology of China, Hefei, China

B.Sc., Chemistry — Sep. 2012 – Jun. 2016
University of Science and Technology of China, Hefei, China

Professional Experience

Doctoral Researcher — Sep. 2021 – Sep. 2026
Scientific Machine Learning Laboratory, Uppsala University
Advisors: Andreas Hellander, Prashant Singh

Machine Learning Engineer Intern — Jan. 2026 – Apr. 2026
Modulai, Stockholm, Sweden

Data Scientist (Part-time) — Jan. 2021 – Jun. 2021
Scaleout Systems AB, Uppsala, Sweden

Research Interests

- Probabilistic and generative modeling
- Uncertainty quantification for multimodal foundation models
- Post-hoc adaptation and representation geometry of pre-trained encoders
- Distributed and federated learning algorithms and systems

Selected Publications

Full publication record available on Google Scholar.

Conference

Ju, L., Nautiyal, M., Vats, E., & Singh, P. (2026). Epistemic Uncertainty Quantification for Pre-trained VLMs via Riemannian Flow Matching. *Proceedings of the 43rd International Conference on Machine Learning (ICML 2026).*

Ju, L., Andersson, M., Fredriksson, S., Glöckner, E., Hellander, A., Vats, E., & Singh, P. (2025). Exploiting the Asymmetric Uncertainty Structure of Pre-trained VLMs on the Unit Hypersphere. *Advances in Neural Information Processing Systems 38 (NeurIPS 2025).*

Ju, L., Singh, P., & Toor, S. (2021). Proactive Autoscaling for Edge Computing Systems with Kubernetes. In *Proceedings of the 14th IEEE/ACM International Conference on Utility and Cloud Computing Companion (UCC 2021).*

Journal

Ju, L., Zhang, T., Toor, S., & Hellander, A. (2024). Accelerating Fair Federated Learning: Adaptive Federated Adam. *IEEE Transactions on Machine Learning in Communications and Networking*, **2**, 1017–1032.

Ju, L., Hellander, A., & Spjuth, O. (2024). Federated Learning for Predicting Compound Mechanism of Action Based on Image Data From Cell Painting. *Artificial Intelligence in the Life Sciences*, **5**.

Under Review

Nautiyal, M., **Ju, L.**, Hellander, A., Vats, E., & Singh, P. (2026). GeoFlowVLM: Geometry-Aware Joint Uncertainty for Frozen Vision-Language Embeddings. *arXiv:2605.13352*.

Nautiyal, M., **Ju, L.**, Ernfors, M., Hagland, K., Holma, V., Söderholm, M. W., Hellander, A., & Singh, P. (2026). OneFlowSBI: One Model, Many Queries for Simulation-Based Inference. *arXiv:2601.22951*.

Funding & Grants

GPU Compute Grants — 2023 – 2025
National Academic Infrastructure for Supercomputing in Sweden (NAISS)
Allocations: 2023/22-1048, 2024/22-1358, and 2025/22-1350
36,000 GPU hours total (12,000 per allocation).

Ph.D. Student Travel Grant — 2021 – 2026
Centre for Interdisciplinary Mathematics, Uppsala University; SEK 50,000.

Honors and Awards

1st Place in Europe, 2nd Place Globally — 2025
Huawei Wireless Communication Global Hackathon, Huawei Sweden and Huawei Global

2nd Place — 2024
Huawei Sweden Hackathon, Huawei Sweden

M.Sc. in Chemometrics, *with Honors* — 2016 – 2019
University of Science and Technology of China

Second-Class Freshman Scholarship — 2012
University of Science and Technology of China

Invited Talks & Presentations

Invited Talks

- Aug. 2026 — "Geometry-Aware Uncertainty Quantification for Frozen Vision–Language Models," GeUmetry Deep Learning Workshop, Sweden.
- Sep. 2025 — "Orthogonality in Neural Networks," Scaleout Edges, Sweden.
- May 2024 — "Federated Learning for Predicting Compound Mechanism of Action Based on Image Data from Cell Painting," AI4Research, Uppsala University.
- Apr. 2024 — "Denoising Diffusion Probabilistic Models," Ph.D. Course on Bayesian Inference, Uppsala University.
- Dec. 2023 — "Federated Learning from the Perspective of Optimization," Centre for Interdisciplinary Mathematics, Uppsala University.

Posters

- Oct. 2024 — "Is Logit Adjustment a Free Lunch for Heterogeneous Federated Learning?" eSSENCE 2024, Sweden.
- Oct. 2022 — "Accelerating Fair Federated Learning," eSSENCE 2022, Sweden.

Teaching & Supervision

Head Teaching Assistant — 2026
Data Engineering II (1TD076), Uppsala University

Teaching Assistant — 2022 – 2025

Data Engineering II (1TD076), Uppsala University

Guest Lecturer 2025
Distributed and Parallel Computing (1TD070), Uppsala University

Teaching Assistant 2021 – 2022
Data Engineering I (1TD169), Uppsala University

Supervisor 2023 – 2026
Supervised four master thesis/course projects on uncertainty quantification, generative modeling, and simulation-based inference, Uppsala University.

Service

Leadership

President of Uppsala Chapter of the Society for Industrial and Applied Mathematics 2024 – 2025
Uppsala University, Uppsala

Academic Reviewing & Program Committees
ICML 2026, NeurIPS 2026, SaTML 2027, IEEE TNNLS, IEEE TMC, IEEE CM.

Open-Source Software

Blades (Contributor) 2022 – 2023
Simulator and benchmark suite for Byzantine-robust federated learning.

FEDn (Contributor) 2020 – 2021
Open-source federated learning framework; contributed PyTorch integration.

Technical Skills

Frameworks	PyTorch, JAX, transformers, verl, vllm, FastAPI, pthread, OpenMP, MPI, CUDA
Cloud / DevOps	Docker, Apptainer, Kubernetes, OpenStack
Programming Languages	Python, C/C++, Haskell, Erlang, Lisp/Racket
Spoken Languages	Chinese (native), English (fluent)

References

Assoc. Prof. Andreas Hellander Ph.D. advisor
Department of Information Technology, Uppsala University
andreas.hellander@it.uu.se

Assoc. Prof. Prashant Singh Co-advisor
Science for Life Laboratory & Department of Information Technology, Uppsala University
prashant.singh@it.uu.se

Assis. Prof. Ekta Vats Collaborator
Department of Information Technology, Uppsala University
ekta.vats@it.uu.se

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